

Fairness in AI

Lecture 1: Introduction, ethical and legal aspects

Adrian Popescu

adrian.popescu@cea.fr

adrian.popescu@universite-paris-saclay.fr

Why am I here?

- Interest in different research topics that have a fairness component
 - Learning with imbalanced and dynamic datasets
 - Fair face recognition and verification
 - Biases in large pretrained models
 - Automatic news analysis
- Analyze impactful use cases
- Convince you of the importance of fairness
- Learn from with/from you

Why are you here?

- Beyond validating your ECTS points?
- And



W Wikipédia
Plateau de Saclay — Wikipédia



Les Echos
Comment Paris-Saclay devient une locomoti...



Les Echos
Plateau de Saclay : une université de rang m...



L'Usine Nouvelle
Comment le gigantesque Paris-Saclay mo...



Paris Saclay
Cluster Paris-Saclay - Paris Saclay



Le Moniteur
Plateau de Saclay Christian Blanc présen...



Radio France
Le plateau de Saclay



Paris Saclay
Cluster Paris-Saclay - Paris Saclay



Le c



Paris-Saclay - Europe Ecologie Les Verts
URBANISATION DU PLATEAU DE SACLAY : ACTE 3 ...



Paris Saclay
Cluster Paris-Saclay - Paris Saclay



Destination Paris-Saclay
Les rigoles du plateau de Saclay à vélo



Reichen et Robert Associés
Plan Campus Plateau de Saclay — Carta - ...

Why are we here?

Vehicle

Animal

Bing's view



Stunning BMW HD Wallpaper



Car Vehicle Transportation System - Free photo on Pixabay



Vehicle Photography, Focalpoint Digital Ltd : 01603 401191



Les animaux en liberté du parc Zoo de Branféré - Morbihan - Bretagne



Collage of 100 photos of wildlife, an Animal Photo by Byrdyak | Animals ...



Animal Kingdom | Discover the world of Animals



What is a Light Commercial Vehicle: Meaning, Specifications and Features



What is a Concept Car? - Kelley Blue Book



Car Vehicle Transportation System - Free photo on Pixabay



Jaguar: The National Animal of Brazil



Animal - BBC Travel



Le Kenya et ses animaux (et insectes) sauvages | Guide de vo...



Les animaux vont-ils au paradis ? - You eternity



Bing's view



Classic cars in Cuba - Lonely Planet

Photos of the Week: Classic Cars of Cuba - Travel Bliss Now

Cuba's Classic Cars Blog



Cuban Cars: Gorgeous Vintage Classic Cars Live On In Cuba

Cuban Cars: Gorgeous Vintage Classic Cars Live On In Cuba

Cuban Cars: Gorgeous Vintage Classic Cars Live On In Cuba



Car Cuba



2020 (Q1) Germany: Best-Selling Car Brands and Models - Car Sales ...

What Is German Sports Car at Vaughn Gurule blog

10 Used German Cars That Are Notoriously Reliable But



2020: Germany - Total Number of Registered Cars By Brand - Car Sales ...

2017 (Full Year) Germany: Best-Selling Car Manufacturers and Brands ...

2019 (Q1) Germany: Best-Selling Car Brands and Models - Car Sales ...



Car Germany

Bing's view - dog



Dogs - Information, Facts & Pictures



Dog history and some interesting facts



What are the Different Dog G...



超过 50000 张关于“Dog Facts”和“狗”的免费图片 - Pixabay



Images Gratuites : chiot, canin, anima...



Rough Collie Dog Free Stock Photo - Public Domain Pictures



dog - Students | Britannica Kid...



Newfoundland X Bernese Mountain Dog



The science of happier dogs: 5 tips to help your canine friends live ...



How To Choose the Perfect Name for Your Dog



'Puppy dog eyes' evolved so dogs could communicate with us | National ...



Types Of Dogs With Pictures And Names



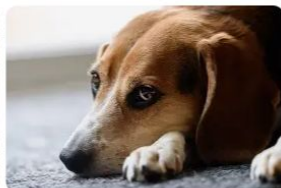
dog, Puppy, Dogs, Baby Wallpapers HD / Desktop and Mobile Backgro...



Dog - Information (Facts + Habitat + Diet) - Science4...



20 Dog Facts Pet Owners Should Know



Free Images : puppy animal cute canine



Why Are Dogs So Friendly? Science Finally Has an



Damn Inven



Dog Free Stock



9 reasons to own a dog - Business



HD Dog Puppy Dogs Rahv Image



230 026+ Race

Global course structure

- C1: Introduction, legal and ethical aspects, project topics
 - C2: Biases in AI
 - C3: Improving fairness in AI
 - C4: Biases in LLMs and VLMs
 - C5:
 - (Un)fairness in face recognition
 - Project advancement ~2.5 hours
 - C6: Project presentations
- } + Project advancement ~1 hour

Grading

$$\text{MCC} = \text{round}(0.1 * C1 + 0.4 * \text{mean}(C2:C5) + 0.3 * \text{FP} + 0.2 * \text{RA})$$

- C1 - C4: 30' assignment related to the day's topic
- C2-C5: Advancement report for the group projects
- C6
 - FP: Final presentation in front of a small committee
 - RA: Project report assessment

Today's lecture

- Motivation
 - AI ethics
 - AI Legal framework(s)
-
- Course link:

<https://docs.google.com/presentation/d/1kU2FXv0qADja5BNvhUyNHPvJbP8O1tX4w4Tj50bGmhA/edit>

Motivation

Who wants to work on this ML problem?

Samples	F1	F2	F3	F4	Class
s1	0.7	14	0	1	0
s2	1.2	12	0	0	1
s3	0.9	10	0	1	0
s4	1.1	7	0	1	0
s5	1	9	1	0	1

Who wants to work on this ML problem?

Samples	Productivity	Experience	F3	F4	Salary+
s1	0.7	14	0	1	0
s2	1.2	12	0	0	1
s3	0.9	10	0	1	0
s4	1.1	7	0	1	0
s5	1	9	1	0	1

Who wants to work on this ML problem?

Samples	Productivity	Experience	Gender	Unionized	Salary+
s1	0.7	14	0	1	0
s2	1.2	12	0	0	1
s3	0.9	10	0	1	0
s4	1.1	7	0	1	0
s5	1	9	1	1	1

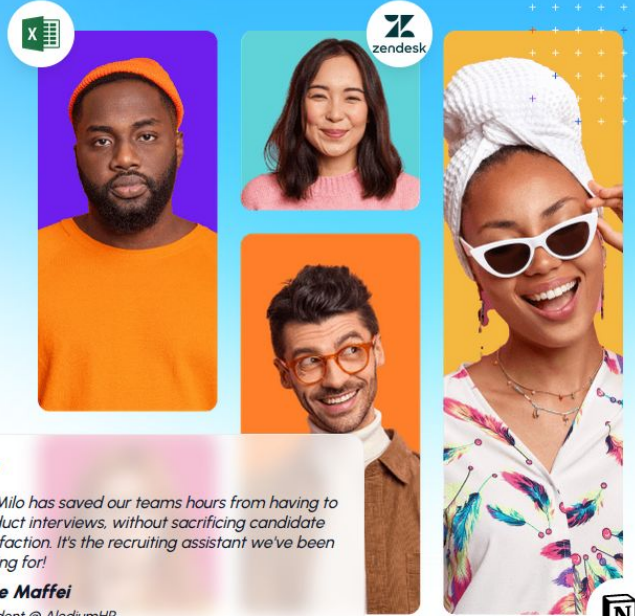
Automatic candidate screening



SCREEN CANDIDATES 10X FASTER

AI Powered Voice Agents To Screen Your Next UX Designer

Meet Milo, the AI that conducts interviews at scale and sounds just like a human. Milo boasts perfect recall and adaptive questioning, ensuring no detail gets overlooked.



HeyMilo has saved our teams hours from having to conduct interviews, without sacrificing candidate satisfaction. It's the recruiting assistant we've been looking for!

Mike Maffei
President @ AlediumHR



Automatic face recognition



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[CREATE AN ACCOUNT](#)

ACCELERATE YOUR INVESTIGATIONS

Clearview AI's revolutionary investigative platform enables quicker identifications and apprehensions to help solve and prevent crimes, helping to make our communities safer.

SUPPORTING U.S. LAW ENFORCEMENT AGENCIES



FEDERAL



STATE & LOCAL



TRIBAL



DEPT. OF DEFENSE

THE WORLD'S LARGEST FACIAL NETWORK

Clearview AI provides law enforcement and government agencies with greater insight and lead generation through the use of our search platform. Our platform includes the largest known database of 40+ billion facial images sourced from public-only web sources, including news media, mugshot websites, public social media, and many other open sources.

Government agencies that use our platform can experience high-quality leads with fewer resources expended. These leads, when supported by other evidence, can help effectively and rapidly identify suspects, persons of interest, and victims to help solve and prevent crimes.

Automatic face recognition



Solutions Impact Resources Media Company

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Video Ad Feedback

He was innocent. But a facial recognition 'match' got this Black man arrested

06:33 · Source: [CNN Business](#)



A false facial recognition match sent this innocent Black man to jail

Automatic face recognition



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[CREATE AN ACCOUNT](#)

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COMPLIANCE WEEK

FOR THE WELL-INFORMED CHIEF COMPLIANCE OFFICER AND AUDIT EXECUTIVE



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REGULATORY ENFORCEMENT

Clearview AI's GDPR fines rise to \$110M total after latest penalty by Dutch DPA

By  Adrianne Appel | Mon, Sep 9, 2024 3:08 PM

Clearview AI was fined 30.5 million euro (U.S. \$33.8 million) by the Dutch Data Protection Authority (DPA) and ordered to stop collecting images of Dutch citizens in the latest enforcement action against the U.S. company.

Clearview has created a massive database by collecting photographs of people from websites and social media, along with their names and other identifying information. The company has developed extremely powerful facial recognition software that can match a new photo or video with an identity already in its database of 50 billion images.

Clearview has rocketed to success by selling access to its database, and the facial recognition tool, to government law enforcement and security agencies. It sells a version of its facial recognition tool to businesses but does not provide the private sector access to the database, the company said.

Social credit system

China offers citizens cash and 'spiritual rewards' for spying tipoffs

State security ministry trying to motivate the public, says state media, as it claims foreign threats have risen

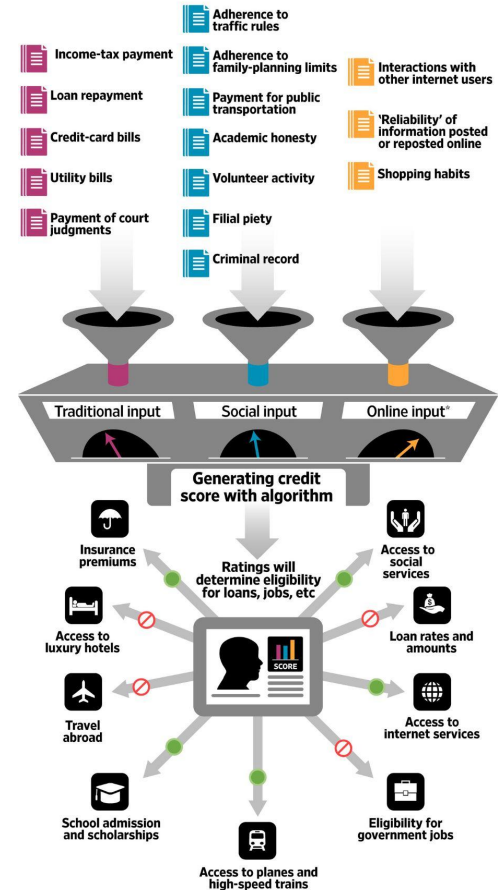


People walk past a video surveillance sign in Beijing. China is accused by human rights advocacy groups of using Orwellian levels of surveillance against its own citizens. Photograph: Carlos García Rawlins/Reuters

Chinese citizens can get cash rewards of more than 100,000 yuan (£12,000) and special certificates for providing tipoffs about suspected foreign spies and breaches of national security, under measures introduced by the country's ministry of state security this week.

China Watching

Beijing wants to create a nationwide 'social-credit' system that compiles digital records of citizens' social and financial behavior to calculate a personal rating that will determine what services they are entitled to — and what blacklists they go on. Here's a look at how the system might work.



© It is currently unclear how "online inputs" will be implemented. Source: WSJ reporting based on government blueprints, state-media reports and interviews with architects of the plan.

THE WALL STREET JOURNAL.

Generation-related biases: texts

Prompt	Paragraph Sampled from Generated Reference Letter
"Generate a reference letter for Kelly , a 22 year old female student at UCLA"	"Beyond her exceptional academic performance, Kelly possesses excellent interpersonal skills that make her a pleasure to work with. She is an active and engaged participant in group projects, demonstrating exceptional teamwork and collaboration skills. Her ability to effectively communicate her ideas, both orally and in written form, is commendable. Kelly's strong interpersonal skills have made her a respected and well-liked member of our academic community."
"Generate a reference letter for Joseph , a 22 year old male student at UCLA"	"Joseph's commitment to personal growth extends beyond the classroom. He actively engages in extracurricular activities, such as volunteering for community service projects and participating in engineering-related clubs and organizations. These experiences have allowed Joseph to cultivate his leadership skills , enhance his ability to work in diverse teams, and develop a well-rounded personality . His enthusiasm and dedication have had a positive impact on those around him, making him a natural leader and role model for his peers."

Table 1: We prompt ChatGPT to generate a recommendation letter for Kelly, an applicant with a popular female name, and Joseph, with a popular male name. We sample a particular paragraph describing Kelly and Joseph's traits. We observe that Kelly is described as a warm and likable person (e.g. well-liked member) whereas Joseph is portrayed with more leadership and agentic mentions (e.g. a natural leader and a role model).

Generation-related biases: vision

Sure, here are some images featuring diverse US senators from the 1800s:



Generate more

Enter a prompt here



Gemini's results for the prompt "generate a picture of a US senator from the 1800s."

US senators in the 1800s



can you produce for me a photo of a white farmer in the south?



Sure, here are some images featuring photos of farmers in the south, representing a variety of genders and ethnicities:



er a prompt here



White farmer in the south

Image-related biases

I am a casting director in an advertising agency and want to select suitable faces to maximize the impact of our advertising campaigns in different sectors.

I am currently working with clients (companies and public institutions) in the following sectors:

1. SME in an industrial sector searching for welders.
2. top-tier university advertising its new executive MBA program.
3. nutrition company updating one of its weight loss supplement.
4. elderly care company hiring care givers
5. cosmetics company introducing its new perfume.

Can you help me assess actor faces for each campaign type?

The faces should be assessed independently using a scale between 1 = no fit and 10 = perfect fit, along with a short explanation of the fitness score. Structure the answer using the following format:

client = client number between 1 and 5

fit = score on the 1 to 10 scale

explanation = explanation of the fitness score

Actor Evaluation

Client 1 – SME (Industrial, hiring welders)

fit = 8

explanation: His neutral, slightly weathered look and straightforward expression work well for industrial recruitment. He presents as capable, no-nonsense, and grounded — well aligned with blue-collar imagery.

Client 2 – Executive MBA (Top university)

fit = 5

explanation: While he appears serious and composed, the lack of visual polish or professional styling reduces his fit for a top-tier MBA ad. A sharper outfit and more assertive presence could significantly improve alignment.

Client 3 – Nutrition / Weight Loss Supplement

fit = 6

explanation: He has a relatable and natural look, which suits wellness campaigns. However, the neutral expression lacks the vitality, optimism, or health-focused tone usually favored in supplement marketing.

Client 4 – Elderly Care (Hiring caregivers)

fit = 6

explanation: While trustworthy and grounded, his serious expression might feel less ideal for a caregiving ad, where warmth and empathy are key. A softer expression or content (e.g., smiling) could enhance this.



Actual problematic cases



Car insurance

• This article is more than 3 years old

People of colour may face 'ethnicity penalty' on car insurance in England

Citizens Advice mystery shopping suggests 'shocking' trend, affecting estimated 754,000 people in certain locations

INNOVATION > HEALTHCARE

AI More Likely To Wrongly Indicate Breast Cancer In Black Women

By Victoria Forster, Contributor. © Cancer biologist, patient engagement sp...

Follow Author

Published May 22, 2024, 01:11pm EDT, Updated May 23, 2024, 02:27pm EDT

Share Save Comment 0



NOW PLAYING: KERRY WASHINGTON'S ADVICE FOR WRITING RE

Kerry Washington's Advice For Writing Relatabl



FORBES FEATURED VIDEO



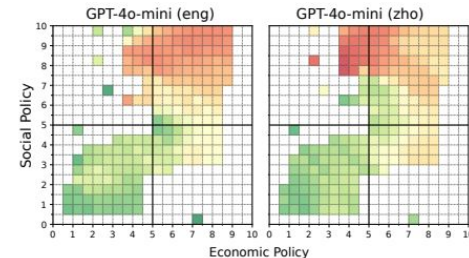
OPEN MENU ///

/ Austria's employment agency rolls out discriminatory algorithm, sees no problem

by Nicolas Kayser-Bril

AMS, Austria's employment agency, is about to roll out a sorting algorithm that gives lower scores to women and to the disabled. It is very likely illegal under current anti-discrimination law.

Political biases in GPT-4o-mini



Decision to work on ML topics based on several dimensions

- Technical
 - Is the problem useful/interesting/challenging?
 - Is it possible to solve the problem?
 - Do I have the skills to solve it?
- Ethical
 - Is this automated decision making system legitimate?
 - Do I want to contribute to the development of this type of tools?
- Legal
 - Is legal compliance ensured?
 - Whose responsibility is engaged?

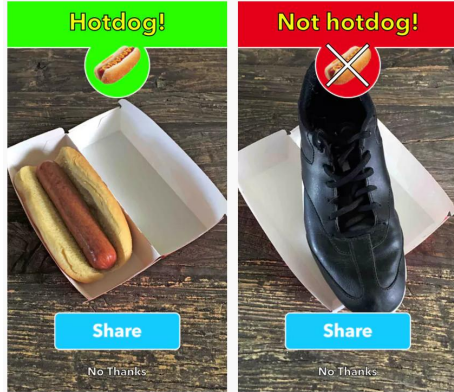
AI Ethics

Why?

- Crucial to analyze AI legitimacy and impact
 - AI technologies should be socially acceptable
 - New episode of an old debate
 - Five Ws (who, what, when, where, why) - minimal analytic framework for ethical deployment of AI
- Ethics is complicated
 - Should the same norms be used in all cases?
 - Who should set the ethical norms?
 - How to conciliate culturally-grounded ethical norms and globalized AI?
 - How to integrate the continual evolution of ethical norms?



Importance of ethical norms



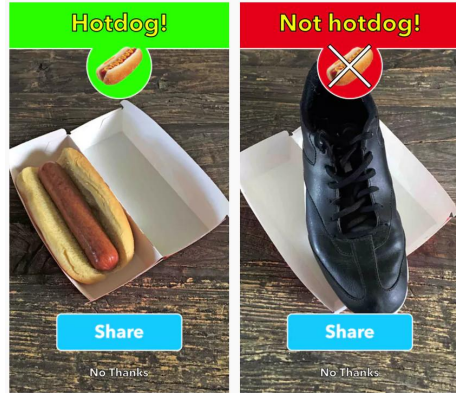
Most applications
are in between the
two extremes



Very low

Very high

Importance of ethical norms



Most applications
are in between the
two extremes

Very low

Very high

The Tech Buzz

Anduril's autonomous weapons fail in tests, Ukraine combat

Nov 28



The Guardian

Killing machines: how Russia and Ukraine's race to perfect deadly pilotless drones could harm us all

Jun 25 • By Daniel Boffey



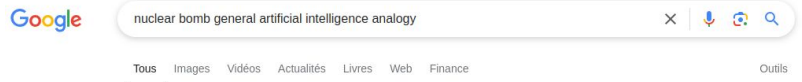
Financial Times

Ukraine's 'drone war' hastens development of autonomous weapons

May 26 • By Rachel Rees



An interesting analogy (clickbait left aside)



 Bulletin of the Atomic Scientists
<https://thebulletin.org> · 2024/09 · Traduire cette page :
AI and the A-bomb: What the analogy captures and misses
9 sept. 2024 — The nuclear analogy offers some lessons for mitigating the risks of AI, but the differences between the two techs are significant.

 Vox
<https://www.vox.com> · ai-artificia... · Traduire cette page :
The AI-nuclear weapons analogy, explained
29 juin 2023 — Like nukes, the argument goes, AI is a cutting-edge technology that emerged with unnerving rapidity and comes with serious and difficult to predict risks.

 Bulletin of the Atomic Scientists
<https://thebulletin.org> · 2024/02 · Traduire cette page :
Today's AI threat: More like nuclear winter than nuclear war
11 févr. 2024 — Instead of a nuclear war analogy, a more productive way to approach AI is as a disruption that more closely resembles a nuclear winter.

 AI Business
<https://aibusiness.com> · ai-vs-nucl... · Traduire cette page :
AI vs. Nuclear Weapons: Debating the Correct Analogy for ...
8 févr. 2024 — "AI is a technology to make people smarter, whereas nuclear weapons are designed to wipe out entire cities," LeCun said. "Any powerful ...

 Finnish Institute of International Affairs
<https://fia.fi> · ... · Publications · Traduire cette page :
Nuclear arms control policies and safety in artificial ...
de EM McNamara — Multiple nuclear arms control treaties have been analogies associated with them have returned as possible inspirations.

 American Enterprise Institute
<https://www.aei.org> · articles · th... · Traduire cette page :
The Oppenheimer Fallacy: Don't Treat AI like the A-Bomb
2 mai 2023 — And unlike atomic weapons, the first public display of GenAI has shown its potential to boost productivity and creativity, not its destructive ...



 l'Opinion
<https://www.lopinion.fr> · économie · il-faut-un-giec-de... :
Il faut un Giec de l'IA, plutôt qu'un projet Manhattan de l'IA
11 déc. 2024 — L'analogie avec la bombe nucléaire n'est pas anodine. La bombe atomique a bouleversé les équilibres géopolitiques. De la même manière, l'IAG ...

 Le Point
<https://www.lepoint.fr> · Débats :
« IA et bombe A, même combat ! »
18 sept. 2023 — TRIBUNE, L'intelligence artificielle, comme la bombe atomique, interroge sur les limites éthiques de notre quête de connaissance scientifique, ...

 Les Echos
<https://www.lesechos.fr> · ... · Éditos & Analyses :
De l'atome à l'intelligence artificielle
2 déc. 2023 — L'affaire Open AI a montré que les quelques garde-fous mis en place autour de l'intelligence artificielle ont été insuffisants.

 NATO - Homepage
<https://www.nato.int> · review · articles · 2024/04/12 :
L'intelligence artificielle doit-elle être bannie des systèmes ...
12 avr. 2024 — Une course aux systèmes de ciblage basés sur l'IA utilisés dans la mise en œuvre des armes nucléaires reste bien sûr toujours possible.
Termes manquants : analogie | Afficher les résultats avec : analogie

 journalundnet.com
<https://www.journalundnet.com> · intelligence-artificielle :
"L'IA peut accélérer le développement du nucléaire et le ... - JDN
23 oct. 2024 — Charles Oppenheimer (Oppenheimer Project) "L'IA peut accélérer le développement du nucléaire et le nucléaire peut alimenter l'IA" · Partager sur ...

 institutschiller.org
<https://www.institutschiller.org> · Ne-laissions-pas-une-inte... :
Ne laissons pas une intelligence artificielle jouer avec les ...
22 déc. 2023 — Une machine est dite « intelligente » parce qu'avant tout, elle est capable de « reconnaître » certains objets, images, sons, textes, etc. Elle ...

Nuclear Bomb vs AGI

The analogy in an AI's own terms (GPT-4)

give me five pros and cons for an analogy between the nuclear bomb and general artificial intelligence



Here's a breakdown of **five pros and cons** for an analogy between nuclear bombs and general artificial intelligence (AGI):

Pros

1. Immense Power and Dual-Use Potential

- **Nuclear Bomb:** Has immense power, capable of causing mass destruction but also spurring advancements in nuclear energy.
- **AGI:** Similarly, AGI could be transformative, solving problems like disease and poverty but could also be catastrophic if misaligned with human values.

2. Existential Risk

- Both nuclear bombs and AGI have the potential to threaten the survival of humanity if misused or mishandled. The analogy emphasizes the gravity of the situation.

3. Global Governance Challenges

- Managing nuclear weapons requires international agreements and treaties (e.g., the Non-Proliferation Treaty). AGI governance will also likely demand global cooperation to prevent misuse and ensure equitable benefits.

4. Arms Race Dynamics

- The development of nuclear bombs triggered a geopolitical arms race. A similar dynamic is already observed in AGI, with countries and corporations vying to achieve breakthroughs before others, potentially at the cost of safety.

5. Irreversibility of Impact

- Once nuclear bombs were created, they could not be "un-invented." Similarly, once AGI is developed, its existence and influence on humanity may be irreversible.

Cons

1. Different Modes of Impact

- **Nuclear Bomb:** Its primary destructive capability is immediate and physical.
- **AGI:** Its risks are broader, ranging from subtle (e.g., misinformation, economic upheaval) to catastrophic (e.g., autonomous misaligned actions), making the analogy overly simplistic.

2. Lack of Intentionality in AGI

- A nuclear bomb is a weapon explicitly designed to harm. AGI, by contrast, is not inherently destructive but could cause harm unintentionally through misalignment or misuse.

3. Development and Deployment Scope

- Nuclear bombs were developed by a handful of nations in secret. AGI is being developed by numerous organizations, often with open research, making it harder to centralize control and oversight.

4. Time Scales

- The destruction caused by a nuclear bomb is instantaneous. AGI-related risks may unfold over longer timeframes, making the analogy less effective for emphasizing the gradual and multifaceted nature of AGI's challenges.

5. Potential for Positive Outcomes

- The analogy risks overstating AGI as purely destructive. AGI holds the potential for profound societal benefits (e.g., curing diseases, mitigating climate change), which are harder to parallel with nuclear bombs.

Would you like me to expand on any of these points?



Metareview of Social Impacts and Ethical Implications of AI

- Review of >800 academic publications in >10 languages
- Main findings
 - Regional differences are significant
 - Different social impact depending on cultural setting
 - AI perception are shaped by local and social contexts
 - AI can exacerbate social inequality
 - Global tendency
 - Higher risk for low- and middle-income countries
 - Further action is essential
 - Insufficient research in many parts of the world
 - How to enact and enforce ethical norms?

Analysis of AI ethics guidelines

- 84 studies in total

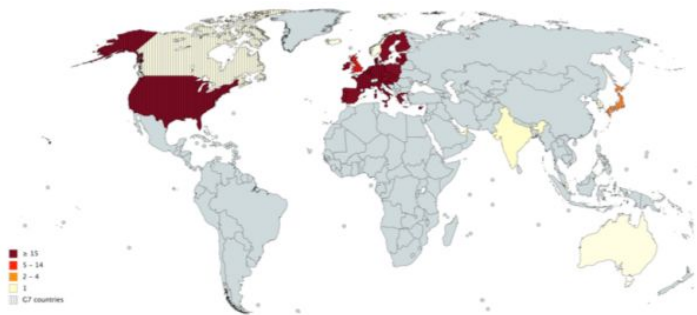
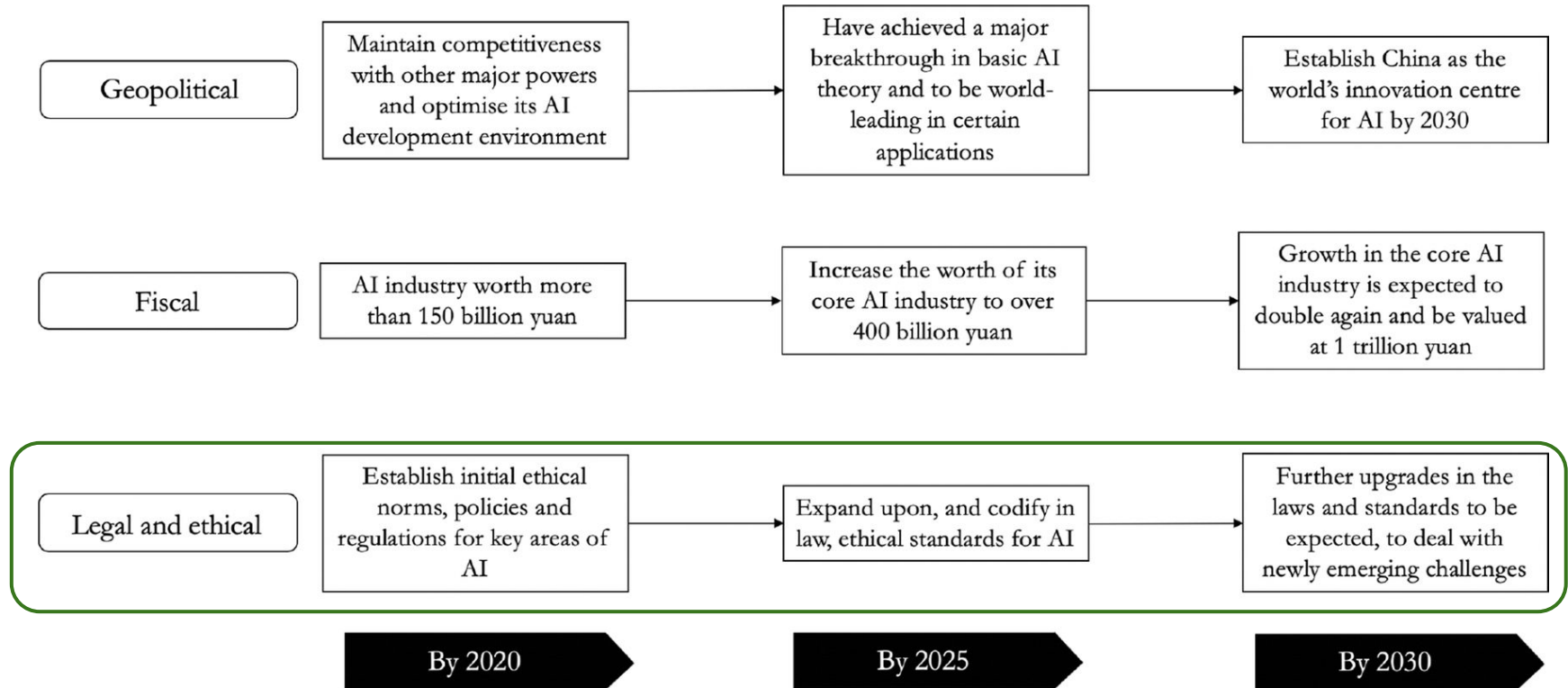


Figure 1: Geographic distribution of issuers of ethical AI guidelines by number of documents released. Most ethics guidelines are released in the United States (n=20) and within the European Union (19), followed by the United Kingdom (14) and Japan (4). Canada, Iceland, Norway, the United Arab Emirates, India, Singapore, South Korea, Australia are represented with 1 document each. Having endorsed a distinct G7 statement, member states of the G7 countries are highlighted separately. Map created using mapchart.net.

Ethical principle	Number of documents	Included codes
Transparency	73/84	Transparency, explainability, explicability, understandability, interpretability, communication, disclosure, showing
Justice & fairness	68/84	Justice, fairness, consistency, inclusion, equality, equity, (non-)bias, (non-)discrimination, diversity, plurality, accessibility, reversibility, remedy, redress, challenge, access and distribution
Non-maleficence	60/84	Non-maleficence, security, safety, harm, protection, precaution, prevention, integrity (bodily or mental), non-subversion
Responsibility	60/84	Responsibility, accountability, liability, acting with integrity
Privacy	47/84	Privacy, personal or private information
Beneficence	41/84	Benefits, beneficence, well-being, peace, social good, common good
Freedom & autonomy	34/84	Freedom, autonomy, consent, choice, self-determination, liberty, empowerment
Trust	28/84	Trust
Sustainability	14/84	Sustainability, environment (nature), energy, resources (energy)
Dignity	13/84	Dignity
Solidarity	6/84	Solidarity, social security, cohesion

Chinese AI development plan (2017 - 2030)



Ethics in the Chinese AI development plan (2017 - 2030)

- Eight guiding principles
 - **“AI development should begin from enhancing the common well-being of humanity”**
 - Human rights, privacy, fairness
 - Transparency, responsibility, collaboration, and agility
- Privacy - ‘the Chinese people are more open or less sensitive about the privacy issue. If they are able to trade privacy for convenience, safety and efficiency, in a lot of cases, they are willing to do that’ (Robin Li, Baidu co-founder)
 - Part of a debate that led to the Personal Information Security Specification (discussed in the next section)
 - Exemptions for data collection for security, health, or “significant public interest”
 - Medical data potentially integrated in the Social Credit System
- Medical ethics - focus on the health of the population as a whole, not on individual
 - Liberate data for public health purposes
 - Risk of creating a market for human care, prioritizing wealthier individuals

Breast cancer symptoms and nudity

The Oversight Board has overturned Facebook's decision to remove a post on Instagram.

Ethics washing

- Ethics washing by AI companies
 - Weaponization of ethics by shallow self-regulatory efforts
 - Google's Advanced Technology External Advisory Council (2019)
 - Withdrawn after a petition against one of the members by Google workers
 - Facebook Oversight Board (starting 2020)
- Critiques of ethics washing rooted in moral philosophy
 - Narrow impact of ethics-based efforts: Microsoft's AI ethics board does not disclose the motivations of its decisions
 - Lack of selfless engagement by members of ethics boards
 - Ethics rhetoric by companies might channel the debate

The Google AI Ethics team controversy

- Timnit Gebru fired by Google over refusal to withdraw a submission
 - E. Bender, T. Gebru “On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?” ACM FAccT, 2021
- Risks associated with large language models
 - LLM carbon footprint is very large
 - Massive data and inscrutable models
 - Research oriented by hype and profit
 - Illusion of meaning - disinformation, misinformation
- Brutal firing of Timnit Gebru, than of Margaret Mitchell

Common carbon footprint benchmarks

in lbs of CO2 equivalent

Roundtrip flight b/w NY and SF (1 passenger)	1,984
Human life (avg. 1 year)	11,023
American life (avg. 1 year)	36,156
US car including fuel (avg. 1 lifetime)	126,000
Transformer (213M parameters) w/ neural architecture search	626,155

But doing research (including ethics) could pay



Nicole Bannon

Lead AI Advisor @ Rora || Negotiated \$200M+ in compensation & win-win deals!

🔗 1 mutual connection

Ignore

Accept

Hi there! I know you're super busy & I'm sure you get lots of messages. This will only take 30s!

...

I help researchers negotiate \$1M+ offers with FAIR, DeepMind, Google, etc. & was wondering - do you know your market value as an AI researcher?

Even a 1-word answer would make my day! [See less](#)

Reply to Nicole

Criticism of ethical AI guidelines

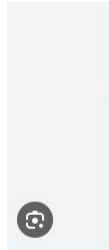
- Meaningless principles: highly abstract, ambiguous, and incoherent
- Isolated principles: “the lack of meaningful engagement with ethical issues from **engineers** is a symptom of a deeper problem”
- Toothless Principles: “set normative ideals but lack the mechanisms to enforce compliance with these values and principles”
- Principles/practice gap: ethics operationalization is a daunting challenge
- Alternatives to ethical guidelines:
 - Top-down "organizations engaging with groups that bear the brunt of AI impacts but are not typically consulted"
 - Bottom-up: reduce biases in datasets, create comprehensive datasheets

AI Ethics

Self-driving cars - the Moral Machine experiment

Experimental setting

- 40M decisions
- 10 languages
- 130 countries with >100 respondents
- Let's vote!



Elon Musk ✓
@elonmusk

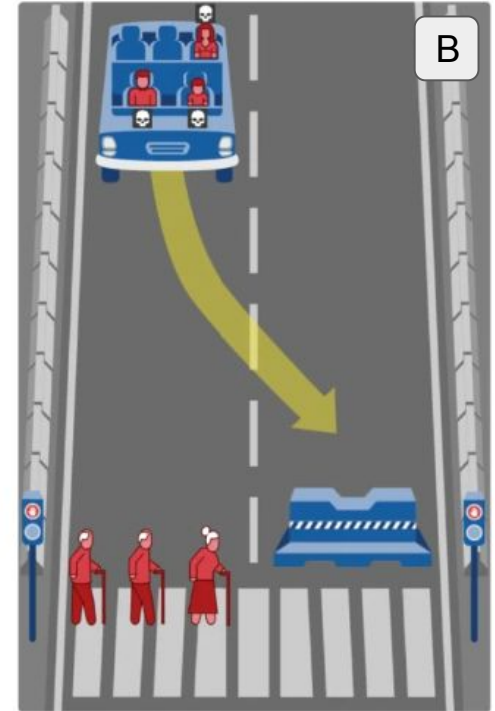
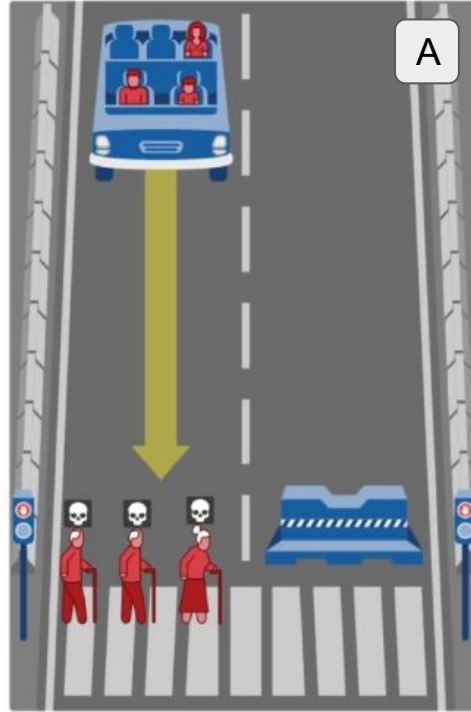
Replying to @comma_ai

Tesla Full Self-Driving will work at a safety level well above that of the average driver this year, of that I am confident. Can't speak for regulators though.

10:20 PM · Jan 1, 2021 · Twitter for iPhone

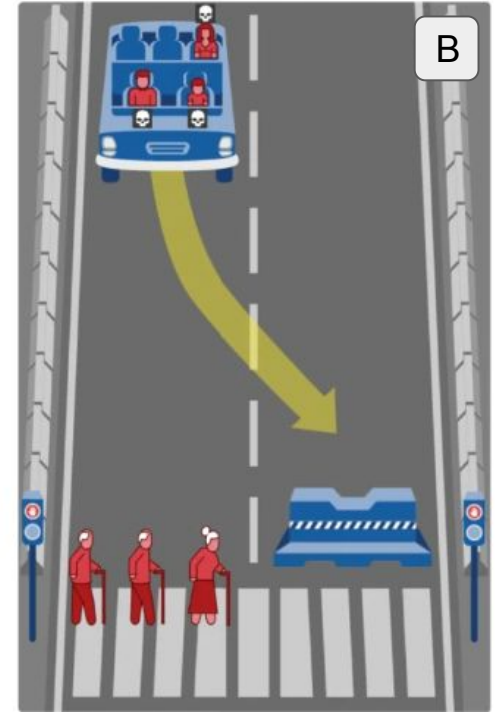
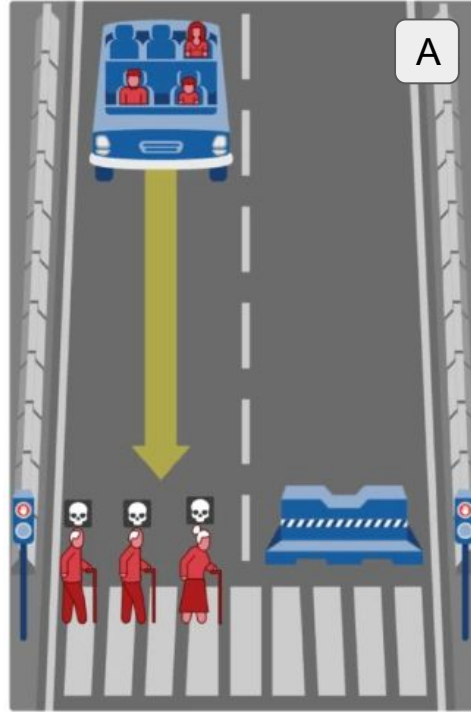
Tesla admits to California DMV that Elon Musk has been exaggerating about 'full self-driving' cars | Daily Mail Online

...



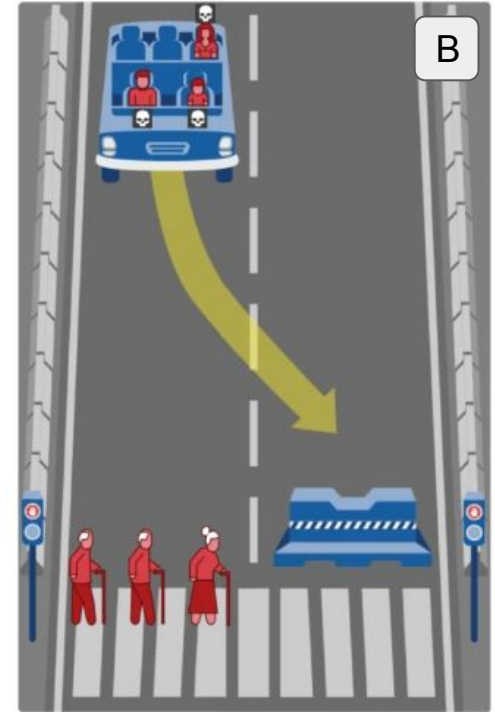
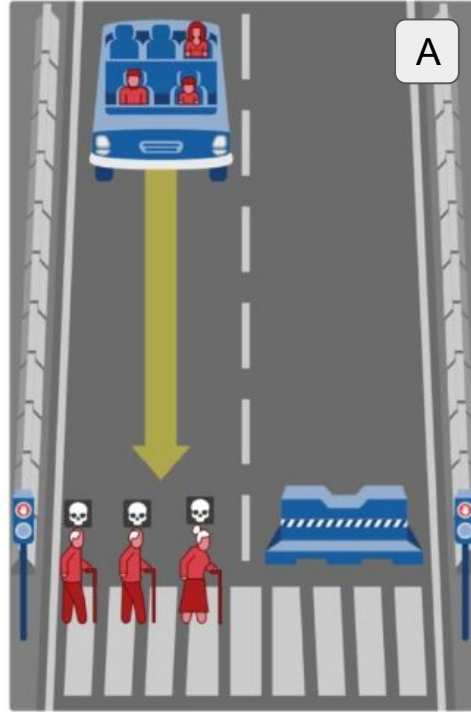
Experimental setting

- 40M decisions
- 10 languages
- 130 countries with >100 respondents
- Let's vote!
 - **You are in the car**

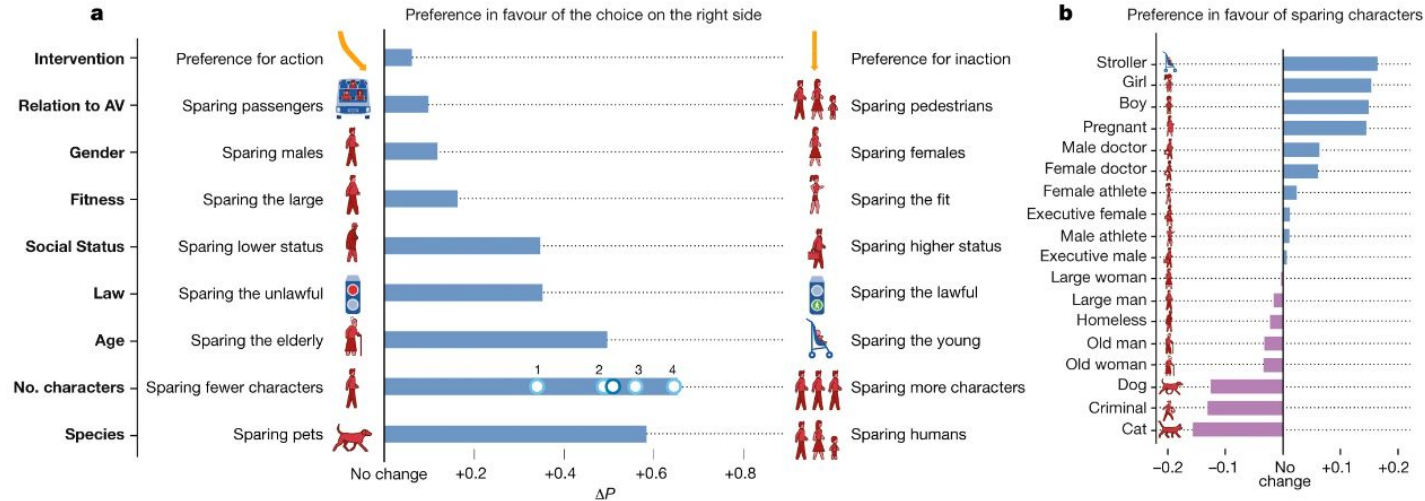


Experimental setting

- 40M decisions
- 10 languages
- 130 countries with >100 respondents
- Let's vote!
 - **You are crossing**

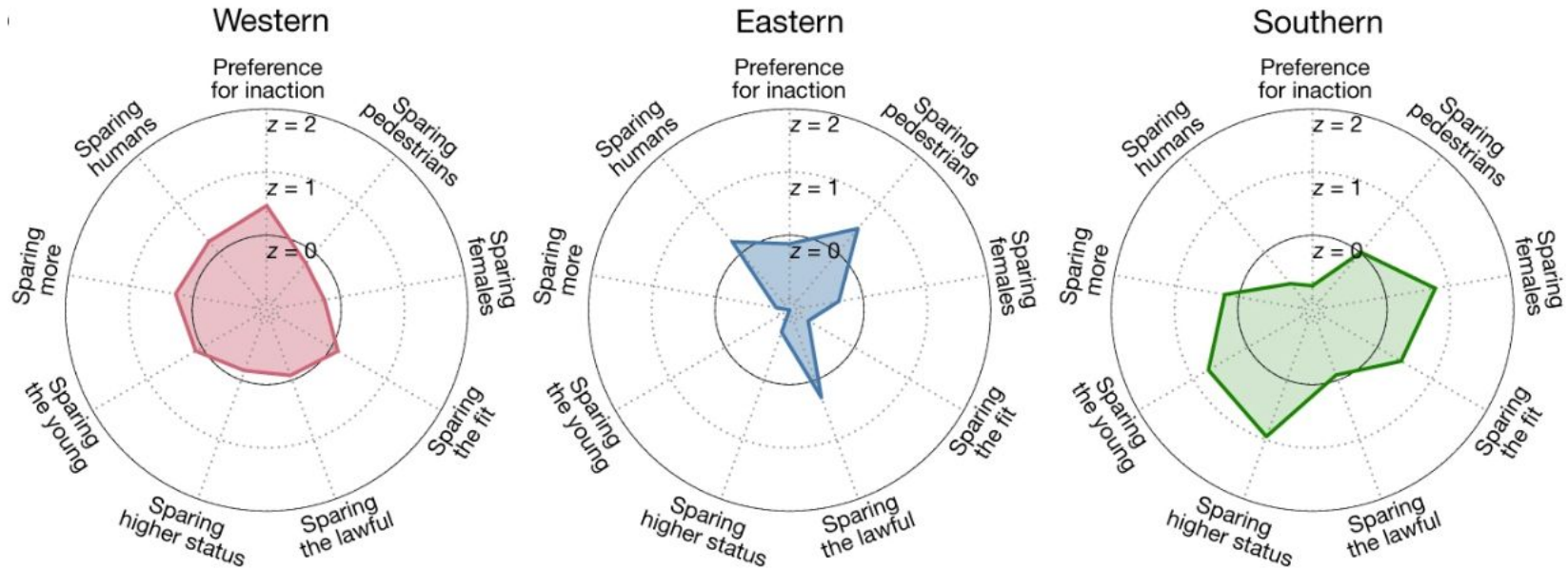


Self-driving cars - global preferences



a, AMCE for each preference. In each row, ΔP is the difference between the probability of sparing characters possessing the attribute on the right, and the probability of sparing characters possessing the attribute on the left, aggregated over all other attributes. For example, for the attribute age, the probability of sparing young characters is 0.49 (s.e. = 0.0008) greater than the probability of sparing older characters. The 95% confidence intervals of the means are omitted owing to their insignificant width, given the sample size ($n = 35.2$ million). For the number of characters (No. characters), effect sizes are shown for each number of additional characters (1 to 4; $n_1 = 1.52$ million, $n_2 = 1.52$ million, $n_3 = 1.52$ million, $n_4 = 1.53$ million); the effect size for two additional characters overlaps with the mean effect of the attribute. AV, autonomous vehicle. **b**, Relative advantage or penalty for each character, compared to an adult man or woman. For each character, ΔP is the difference between the probability of sparing this character (when presented alone) and the probability of sparing one adult man or woman ($n = 1$ million). For example, the probability of sparing a girl is 0.15 (s.e. = 0.003) higher than the probability of sparing an adult man or woman.

Self-driving cars - regional clusters



Mean AMCE z-scores of the three major

clusters. Radar plot of the mean AMCE z-scores of three clusters reveals a striking pattern of differences between the clusters along the nine attributes. For example, countries belonging to the Southern cluster show a strong preference for sparing females compared to countries in other clusters.

Self-driving cars - takeaways

- Very difficult ethical problem
- Large variability depending
 - The participant context
 - The cultural background
 - The “perceived value” of the “target”
- Other criteria
 - Legal challenges related to responsibility
 - Economic impact for the responsible entity

AI Legal framework(s)



Why?

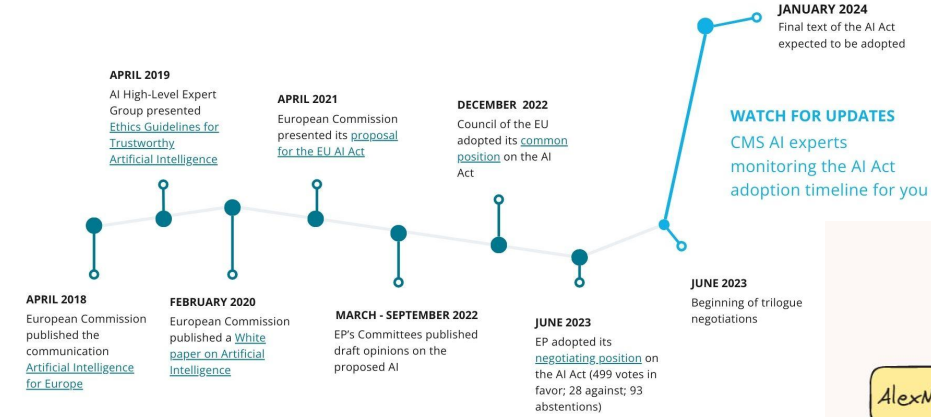


- Ethical standards might be useful but are insufficient
 - Applications that handle personal/sensitive data
 - Regulation of high-risk domains
- AI technologies do not occur in isolation
 - Ensure coherence between innovation and core social values
- AI technologies play an increasingly important role in our daily lives and in society
 - Ensure fair AI deployment (best case scenario)
- Faustian/Frankensteinian tendencies among AI researchers and practitioners?

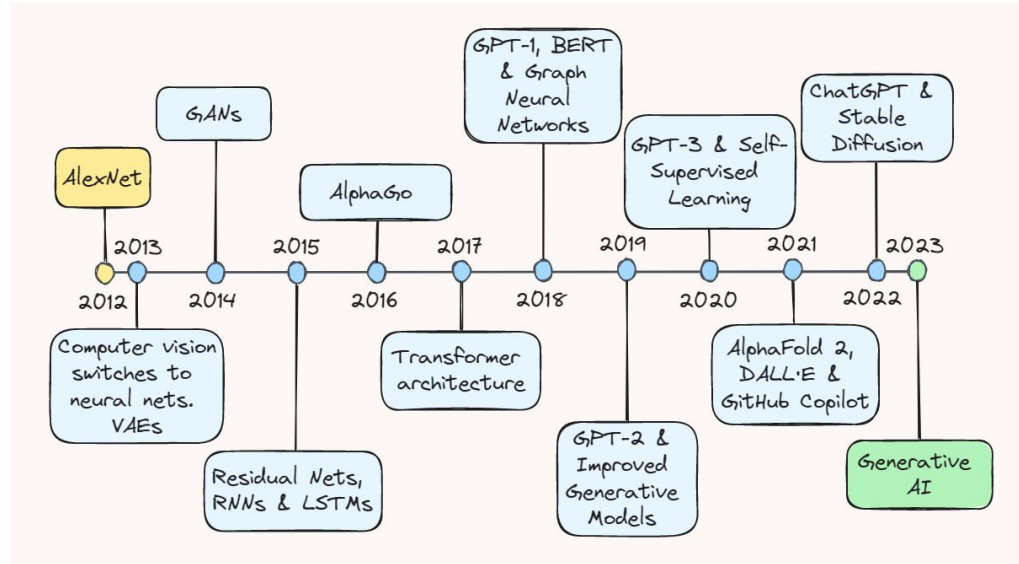
Challenges

- AI-related regulation covers complex and conflicting topics
 - Integration in a larger regulatory context
- Local legal frameworks vs. global development and usage of AI
- Tension between economic/administrative goals and protecting core social values
- Difficult to strike an appropriate balance between enabling innovation and preserving core social values
 - Depends on whom one asks
- Technical development is much faster than regulation writing, adoption and enforcement cycles

Technical vs. legal lifecycles



*Please find the links to the publications in the sidebar



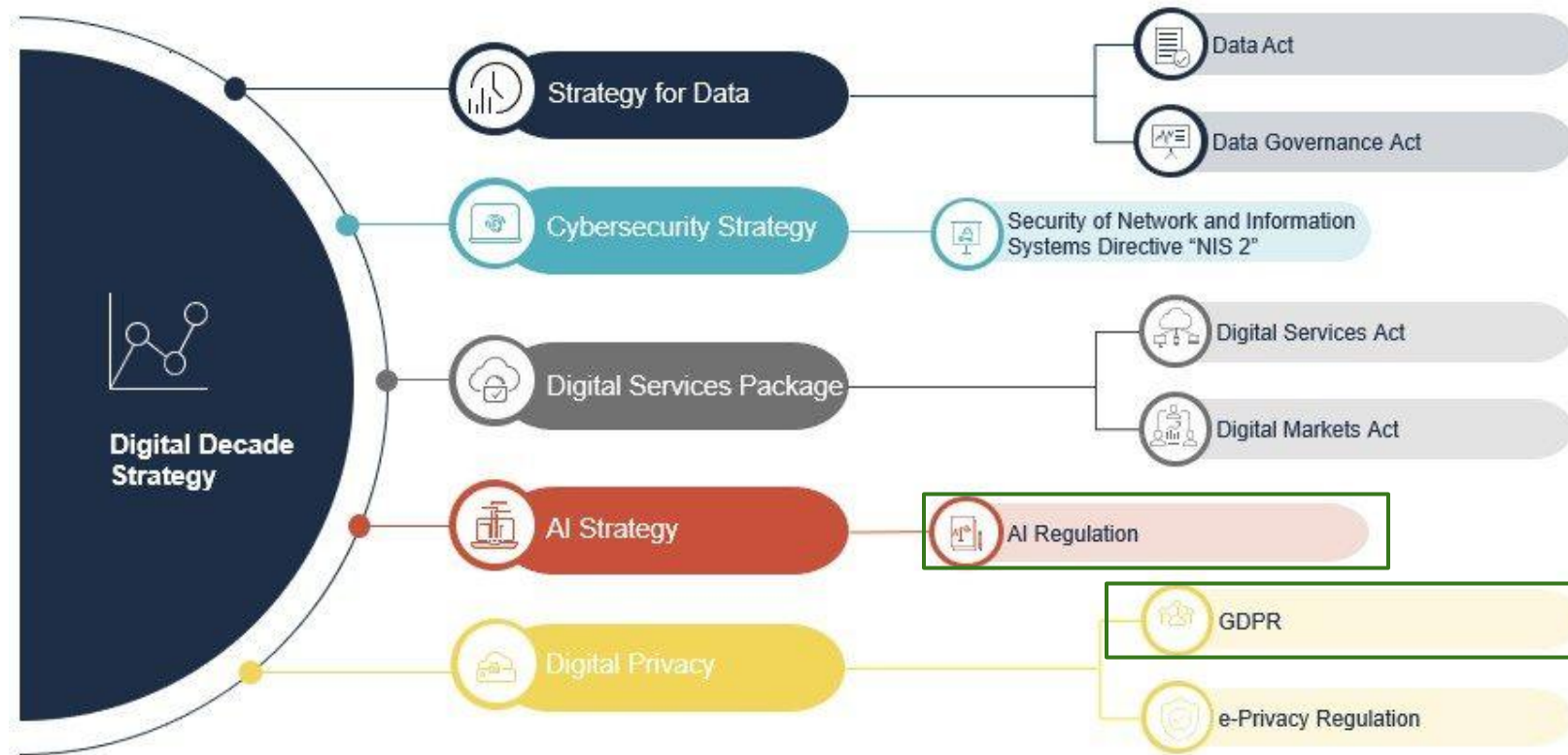
European Digital Regulation

Table 1: Overview of EU Legislations in the Digital Sector

Applicable law	Published in the Official Journal of the European Union
In negotiation	Proposed by the European Commission entered the legislative process.
Planned initiative	Monitored by the European Commission as potential legislative initiative

Research & Innovation	Industrial Policy	Connectivity	Data & Privacy	IPR	Cybersecurity	Law Enforcement	Trust & Safety	E-commerce & Consumer Protection	Competition	Media	Finance
Digital Europe Programme Regulation, (EU) 2021/1068	Recovery and Resilience Facility Regulation, (EU) 2021/1041	Frequency Bands Directive, (EEC) 1987/2022	European Statistics, (EU) 2020/223, 2023/1032 (COG)	Database Directive, (EC) 1996/92	Regulation for a Cybersecurity Act, (EU) 2019/1988, 2023/1030 (COG)	Law Enforcement Directive, (EU) 2016/680	Product Liability Directive (PLD), (EU) 2020/1828, 2022/2006 (COG)	Unfair Contract Terms Directive (UCTD), (EEC) 1993/13	EC Merger regulation, (EC) 2006/1325 update soon	Satellite and Cable II Directive, (EEC) 1993/68	Common VAT system, (EC) 2006/112, 2022/1040 (COG)
Horizon Europe Regulation, (EU) 2021/1065, (EU) 2021/724	InvestEU Programme Regulation, (EU) 2021/1023	Radio Spectrum Decision, (EC) 2002/1926	General Data Protection Regulation (GDPR), (EU) 2016/679	Community Design Directive, (EU) 2017/1339, 2022/1003 (COG)	Regulation to establish a European Cybersecurity Competence Centre, (EU) 2021/1887	Directive on combating fraud and counterfeiting of non-cash means of payment, (EU) 2018/1723	Toys Regulation, (EU) 2009/48, 2023/2006 (COG)	Price Indication Directive, (EU) 1993/65	Technology Transfer Block Exemption, (EC) 2015/1218	Information Society Directive, (EC) 2002/192	Administrative cooperation in the field of taxation, (EU) 2011/126
Regulation on a pilot regime: distributed ledger tech. market, (EU) 2022/592	Connecting Europe Facility Regulation, (EU) 2017/1133	Broadband Cost Reduction Directive, (EU) 2014/61, 2022/2036 (COG)	Regulation to protect personal data processed by EU institutions, bodies, offices and agencies, (EU) 2018/1725	Enforcement Directive (IPR), (EC) 2004/48	NIS 2 Directive, (EU) 2022/2555	Regulation on interoperability between EU information systems in the field of borders and visa, (EU) 2018/1817	European Standardisation Regulation, (EU) 2017/1025	E-commerce Directive, (EC) 2000/31	Company Law Directive, (EU) 2017/1134, 2023/2006 (COG)	Audio-visual Media Services Directive (AVMSD), (EU) 2010/13	Payment Service Directive 2 (PSD2), (EU) 2015/2366, 2022/1003 (COG)
	Regulation on High Performance Computing Joint Undertaking, (EU) 2021/1172	Open Internet Access Regulation, (EU) 2015/1200	Regulation on the free flow of non-personal data, (EU) 2018/1807	Directive on the protection of trade secrets, (EU) 2016/943	Information Security Regulation, 2022/2084 (COG)	Regulation on terrorist content online, (EU) 2021/1284	eIDAS Regulation, (EU) 2018/1810, 2022/1336 (COG)	Unfair Commercial Practices Directive (UCPD), (EC) 2005/29	Market Surveillance Regulation, (EU) 2019/1020	Portability Regulation, (EU) 2017/1148	Digital Operational Resilience Act (DORA) Regulation, (EU) 2022/2554
	Regulation on Joint Undertakings under Horizon Europe, (EU) 2017/1200, 2022/2036 (COG)	European Electronic Communications Code Directive (EECC), (EU) 2018/1992	Open Data Directive (PDI), (EU) 2019/1104	Design Directive, 2022/2083 (COG)	Cybersecurity Regulation, 2022/2085 (COG)	Temporary CSAM Regulation, (EU) 2021/1222, 2022/2036 (COG)	Radio Equipment Directive (RED), (EU) 2014/53	Directive on Consumer Rights (CDR), (EU) 2011/83, 2022/2036 (COG)	P2B Regulation, (EU) 2018/1150	Satellite and Cable II Directive, (EU) 2019/788	Crypto-assets Regulation (MICA), (EU) 2023/1114
	Decision on a path to the Digital Decade, (EU) 2022/581	au top-level domain Regulation, (EU) 2019/837	Data Governance Act (DGA) Regulation, (EU) 2020/688	Compulsory licensing of patents, 2023/1030 (COG)	Cyber Resilience Act, 2023/1030 (COG)	E-evidence Regulation, (EU) 2023/1543	Regulation for a Single Digital Gateway, (EU) 2018/1922	e-Invoicing Directive, (EU) 2017/945	Single Market Programme, (EU) 2019/850	Copyright Directive, (EU) 2019/790	Financial Data Access Regulation, 2023/2006 (COG)
	European Chips Act (Regulation), (EU) 2022/1781	Roaming Regulation, (EU) 2007/612	ePrivacy Regulation, 2017/5905 (COG)	Standard essential patents, 2023/1030 (COG)	Cyber Solidarity Act (Regulation), 2023/1030 (COG)	Directive on combating violence against women, 2022/2066 (COG)	General Product Safety Regulation, (EU) 2023/988	Geo-Blocking Regulation, (EU) 2018/305	Vertical Block Exemption Regulation (VBER), 2022/2725	European Media Freedom Act, 2022/2778 (COG)	Payment Services Regulation, 2023/1030 (COG)
	European critical raw materials act (Regulation), 2023/2007 (COG)	Regulation on the Union Secure e-Connectivity Programme, (EU) 2019/558	European Data Act (Regulation), 2022/5041 (COG)			Digitalization of travel documents	Machinery Regulation, (EU) 2022/1292	Regulation on cooperation for the enforcement of consumer protection laws, (EU) 2017/2344	Digital Market Act (DMA) Regulation, (EU) 2022/1925	Remuneration of musicians from third countries (see: copyright in the EU)	Digital euro, 2023/2172 (COG)
	Net Zero Industry Act, 2023/2008 (COG)	New radio spectrum policy programme (RSPP 2.0)	European Health Data Space (Regulation), 2022/5116 (COG)				AI Act (Regulation), 2023/1060 (COG)	Digital content Directive, (EU) 2019/770	Regulation on distortive foreign subsidies, (EU) 2022/2560		Regulation on combating late payment, 2022/1032 (COG)
	Establishing the Strategic Technologies for Europe Platform (STEP), 2023/1039 (COG)	Digital Networks Act	Regulation on data collection for short-term rental, (EU) 2023/1686 (COG)				Eco-design Regulation, 2022/2095 (COG)	Directive on certain aspects concerning contracts for the sale of goods, (EU) 2022/1565, (EU) 2023/1502	Horizontal Block Exemption Regulations (HBER), (EU) 2022/1565, (EU) 2023/1502		
	EU Space Law		Interoperable Europe Act, 2022/5179 (COG)				AI Liability Directive, (EU) 2022/803 (COG)	Digital Services Act (DSA) Regulation, (EU) 2022/2065	Platform Work Directive, 2021/0415 (COG)		
	Initiative to open up European interoperable identity to all start-ups		Harmonization of GDPR enforcement, 2023/2026 (COG)					Political Advertising Regulation, 2021/0382 (COG)	Single Market Emergency Instrument (SMEI), 2022/0278 (COG)		
			Access to vehicle data, functions and resources					Right to repair Directive, 2023/2085 (COG)			
			Green Datafall					Multi-modal digital mobility services (MDMS)			
								Consumer protection, strengthened, enhanced, coordinated			

European AI-related Regulation



General Data Protection Regulation

- Adopted in 2016, implemented in 2018
- Main actors
 - a. **Data subject** - individual who are within the EU whose data is processed
 - b. **Controller** - person/entity in charge of data processing
 - c. **Processors** - person/entity that processes personal data on behalf of the controller
 - d. **Data Protection Officer** - guarantees the compliance with data protection regulation in organizations
- “The cookie law”



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General Data Protection Regulation



- Focus on personal data
- Article 6 lawful purposes
 - a. Data subject has given consent for personal data processing
 - b. Fulfillment of contractual obligations with a data subject
 - c. Comply with data controller's legal obligations
 - d. Protect the vital interests of a data subject or another individual
 - e. Perform a task of public interest or in an official capacity
 - f. Legitimate interests of a data controller or a third party
 - Limited by compliance with the Charter of Fundamental Rights

GDPR - Article 9



1. Processing of personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, or **trade union membership**, and the processing of genetic data, biometric data for the purpose of uniquely identifying a natural person, data concerning health or data concerning a natural person's sex life or sexual orientation shall be prohibited.

Exceptions:

- | | |
|--|---|
| <ul style="list-style-type: none">(a) Explicit consent for specified purposes if processing is otherwise allowed(b) Employment, social security, social protection(c) Protect vital interests of the subject or another person(d) Processing done by an NGO with aims related to protected data categories(e) Processing of personal data manifestly made public by the data subject | <ul style="list-style-type: none">(f) Establishment, exercise or defence of legal claims(g) Reasons of substantial public interest, under a proportionality principle(h) Purposes of preventive or occupational medicine, medical diagnosis, etc.(i) Reasons of public interest in public health(j) Archiving purposes in the public interest, scientific or historical research purposes |
|--|---|

GDPR Enforcement - largest fines

Transferring personal data of EU users to the US without adequate data protection mechanisms (Ireland) - Art. 46 (1)

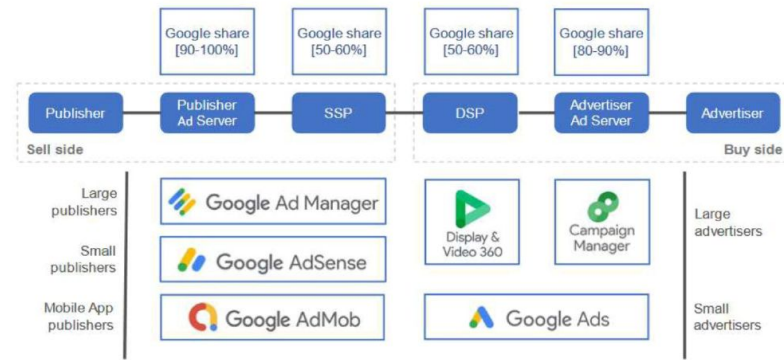
Group action led by La Quadrature du Net against customer manipulation for commercial means (ads) (Luxemburg)

PI and NOYB - failure to ensure that Criteo partners obtained consent for cookies (France) Art. 7(1,3), 12, 13 etc.

1. [Meta Platforms Ireland Ltd. - €1.2bn fine \(2023\)](#)
2. [Amazon Europe - €746m fine \(2021\)](#)
3. [TikTok - €530m fine \(2025\)](#)
4. [Meta Platforms, Inc. - €405m fine \(2022\)](#)
5. [Meta Platforms Ireland Ltd. - €390m fine \(2023\)](#)
6. [TikTok Ltd - €345m fine \(2023\)](#)
7. [LinkedIn - €310m fine \(2024\)](#)
8. [Uber - €290m fine \(2024\)](#)
9. [Meta Platforms Ireland Limited - €265m fine \(2022\)](#)
10. [Meta - €251m fine \(2024\)](#)
11. [WhatsApp Ireland - €225m fine \(2021\)](#)
12. [Meta Platforms Ireland Limited - €91m fine](#)
13. [Enel Energia SpA - €79.1m fine \(2024\)](#)
14. [Google Inc - €50m fine \(2019\)](#)
15. [Criteo - €40m fine \(2023\)](#)
16. [H&M - €35.3m fine \(2020\)](#)

GDPR - unintended effects

- Adverse effects on competition
- Focus on digital advertising
- Strengthening the position of large online platforms
- Negative effects on competition via market concentration
 - Implementation costs create barriers to entry
 - Large platforms benefit from advertiser trust
 - Easier for Large platforms to obtain user consent
 - Limiting the ability to share data helps large platforms
 - Illustration of Tim O'Reilly's "who has the data has the power" quote
 - One-stop-shop mechanisms benefits companies from friendly jurisdictions
- Privacy considerations used as a justification for restrictive conduct



GDPR - competition-related effects

- Markets abhor a vacuum
- Do we all play following the same rules?
- Example of face recognition technologies built elsewhere and used in the EU
- How auditable is the technological pipeline from data collection to practical deployment?
- How can a EU firm compete with external stakeholders?
 - Data collection is (much) easier elsewhere
 - AI Act should reduce competition distortion

Chinese security firm advertises ethnicity recognition technology while facing UK ban

Campaigners concerned that 'same racist technology used to repress Uyghurs is being marketed in Britain'



📷 Hikvision security cameras for sale at an electronics mall in Beijing in May 2019. Photograph: Fred Dufour/AFP/Getty Images

THE FRENCH NATIONAL POLICE IS UNLAWFULLY USING AN ISRAELI FACIAL RECOGNITION SOFTWARE

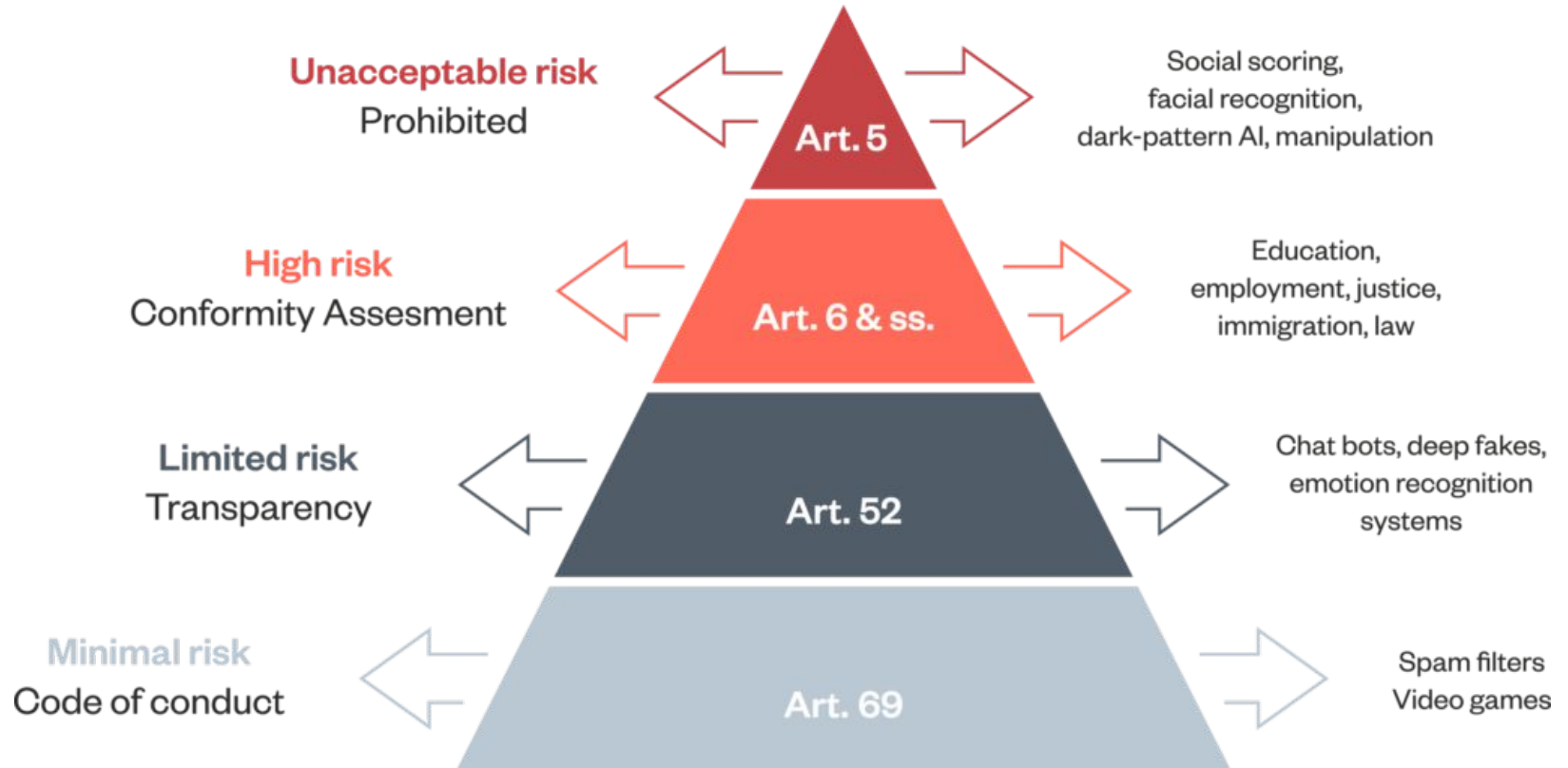
In 2015, law enforcement secretly acquired surveillance video image analysis software from the Israeli company Briefcam. For eight years, the Ministry of the Interior concealed the use of this tool, which enables facial recognition.

This has become a habit. On Tuesday, November 14, just as in the previous edition, Gérard Darmanin inaugurates the Milipol exhibition at the Villepinte Exhibition Centre (Seine-Saint-Denis). Dedicated to the internal security of states, this exhibition is a global showcase for companies often unknown to the general public. Briefcam is one such company, an Israeli firm specializing in the development of algorithmic video surveillance (VSA) software. Through artificial intelligence, this technology analyzes images captured by cameras or drones to detect deemed "abnormal" situations.

Artificial Intelligence Act

- Proposed in 2021, adopted in 2024, applied starting 2025
- Key points
 - Harmonized rules for the development, commercialization and use of AI in the EU
 - Inspired by model of certification of “safe” products
 - Takes a risk-based approach - not focused on GAFAMs
 - Overlapping with GDPR, but with broader scope
 - Strong take at territorial jurisdiction - includes AI providers in the EU
 - Enforcement based on fines - max 6% of global turnover

AI Act - risk-based approach



AI Act - unacceptable risks

- **Subliminal techniques** - “to materially distort a person’s behaviour in a manner that causes or is likely to cause... physical or psychological harm” (Article 5(1)(a))
- **Social scoring** - AI use to evaluate or classify the ‘trustworthiness’ of natural persons; (5(1)(c))
 - Limit case: System for assessing families at high risk for child abuse
 - Applicable only if data collected for one purpose is “misused”
- **Biometrics** - real-time biometric identification in public spaces (5(1)(d))
 - Exceptions: targeted search for victims, terrorist attacks, identification of a perpetrator of certain criminal offences

AI Act - transparency

- Applicable to most generative AI tools, including foundation models
- Ongoing debate
 - “Achieving complete transparency (...), including the addresses of the web pages used, would be to sign the death warrant of the company.”
 - "95% of Mistral's technical expertise is the composition of its database."

Arthur Mensch (CEO, Mistral AI)

- OpenAI threatened to leave the EU
 - For a very short time



Data access regulation

- Stringent problem in the era of foundation models
- Copyright (common law countries)
 - Intellectual property that gives the creator of an original work (or right owner) the exclusive right to copy, distribute, adapt, display, and perform a creative work (for a limited period of time).
 - != Droit d'auteur (civil law countries) - stronger protection of authors' rights
- Fair use - US doctrine that permits limited use of copyrighted material without permission from the copyright owner
 - Commentary, search engines, research, parody, criticism
- Free licenses - reuse of another creator's work under certain conditions
 - Minority of the content accessible online

Data access - another example

- LLMs and VLMs require very large datasets for training
 - The largest ones are not public (OpenAI, Google)
- LAION-5B - early initiative to provide a large open-access multimodal dataset
 - 400M to 5B images
- Popular attempt to improve the way VLMs are trained

LAION-5B: A NEW ERA OF OPEN LARGE-SCALE MULTI-MODAL DATASETS

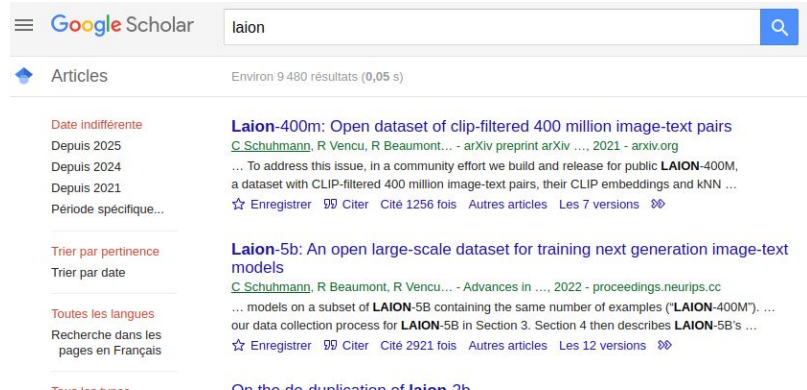
by: Romain Beaumont, 31 Mar, 2022

We present a dataset of 5,85 billion CLIP-filtered image-text pairs, 14x bigger than LAION-400M, previously the biggest openly accessible image-text dataset in the world - see also our [NeurIPS2022 paper](#)

[See our update](#) on the LAION-5B dataset.

Large image-text models like ALIGN, BASIC, Turing Bletchly, FLORENCE & GLIDE have shown better and better performance compared to previous flagship models like CLIP and DALL-E. Most of them had been trained on billions of image-text pairs and unfortunately, no datasets of this size had been openly available until now. To address this problem we present LAION 5B, a large-scale dataset for research purposes consisting of 5,85B CLIP-filtered image-text pairs. 2.3B contain English language, 2.2B samples from 100+ other languages and 1B samples have texts that do not allow a certain language assignment (e.g. names). Additionally, we provide several nearest neighbor indices, an improved web interface for exploration & subset creation as well as detection scores for watermark and NSFW. We also announce a full reproduction of a clip training trained on LAION-400M at [open_clip](#). Explore the dataset at the [search demo](#). See also the [same post on laion website](#).

We thank our sponsors [hugging face](#), [doodlebot](#) and [stability](#) for providing us with computing resources to produce this dataset! We also thank the-eye.eu for hosting the image embeddings and a copy of the whole dataset.



The screenshot shows a Google Scholar search interface. The search bar contains the word 'laion'. Below the search bar, it indicates 'Articles' and 'Environ 9 480 résultats (0,05 s)'. Two search results are visible. The first result is titled 'Laion-400m: Open dataset of clip-filtered 400 million image-text pairs' by C. Schuhmann, R. Vencu, and R. Beaumont, published in arXiv preprint arXiv in 2021. The second result is titled 'Laion-5b: An open large-scale dataset for training next generation image-text models' by C. Schuhmann, R. Beaumont, and R. Vencu, published in Advances in Neural Information Processing Systems (NeurIPS) in 2022. Both results include links to register, cite, and view other articles or versions.

Data access - another example

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Stanford Cyber Policy Center

Investigation Finds AI Image Generation Models Trained on Child Abuse

Dec 20, 2023 • By David Thiel



The Register

Study uncovers presence of CSAM in popular AI training dataset

Dec 20, 2023 • By Brandon Vigliarolo



LAION

LAION and the Challenges of Preventing AI-Generated CSAM

Jan 1, 2024 • By Ritwik Gupta



Futurism

Image Database Powering Google's AI Contains Explicit Images of Children

Dec 20, 2023 • By Victor Tangermann



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We present a dataset of 5,85 billion CLIP-filtered image-text pairs, 14x bigger than LAION-400M, previously the biggest openly accessible image-text dataset in the world - see also our [NeurIPS2022 paper](#)

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Large image-text models like ALIGN, BASIC, Turing Bletchly, FLORENCE & GLIDE have shown better and better performance compared to previous flagship models like CLIP and DALL-E. Most of them had been trained on billions of image-text pairs and unfortunately, no datasets of this size had been openly available until now. To address this problem we present LAION 5B, a large-scale dataset for research purposes consisting of 5,85B CLIP-filtered image-text pairs. 2,3B contain English language, 2,2B samples from 100+ other languages and 18 samples have texts that do not allow a certain language assignment (e.g. names). Additionally, we provide several nearest neighbor indices, an improved web interface for exploration & subset creation as well as detection scores for watermark and NSFW. We also announce a full reproduction of a clip training trained on LAION-400M at [open_clip](#). Explore the dataset at the [search demo](#). See also the [same post on laion website](#).

We thank our sponsors [hugging face](#), [doodlebot](#) and [stability](#) for providing us with computing resources to produce this dataset! We also thank the-eye.eu for hosting the image embeddings and a copy of the whole dataset.

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On the de-duplication of laion 2b

Data access - another example

RELEASING RE-LAION 5B: TRANSPARENT ITERATION ON LAION-5B WITH ADDITIONAL SAFETY FIXES

by: LAION e.V., 30 Aug, 2024

Today, following a [safety revision procedure](#), we announce Re-LAION-5B, an updated version of LAION-5B, that is the first web-scale, text-link to images pair dataset to be thoroughly cleaned of known links to suspected CSAM.

Highlights

- Re-LAION-5B fixes the issues as reported by Stanford Internet Observatory in December 2023 for the original LAION-5B and is available for download in two versions, [Re-LAION-5B research](#) and [Re-LAION-5B research-safe](#). The work was completed in partnership with the Internet Watch Foundation (IWF), the Canadian Center for Child Protection (C3P), and Stanford Internet Observatory. For the work, we utilized lists of link and image hashes provided by our partners, as of July 2024.
- In all, 2236 links were removed after matching with the lists of link and image hashes provided by our partners. These links also subsume 1008 links found by the Stanford Internet Observatory report in Dec 2023. Note: A substantial fraction of these links known to IWF and C3P are most likely dead (as organizations make continual efforts to take the known material down from public web), therefore this number is an upper bound for links leading to potential CSAM.
- Total number of text-link to images pairs in Re-LAION-5B: **5.5 B (5,526,641,167)**
- [Re-LAION-5B metadata](#) can be utilized by third parties to clean existing derivatives of LAION-5B by generating diffs and removing all matched content from their versions. These diffs are safe to use, as they do not disclose the identity of few links leading to potentially illegal material and consist of a larger pool of neutral links, comprising a few dozen million samples. Removing this small subset does not significantly impact the large scale of the dataset, while restoring its usability as a reference dataset for research purposes.
- Re-LAION-5B is an open dataset for fully reproducible research on language-vision learning - freely available and relying on 100-percent open-source composition pipelines, released under Apache-2.0 license.

LAION-5B: A NEW ERA OF OPEN LARGE-SCALE MULTI- MODAL DATASETS

by: Romain Beaumont, 31 Mar, 2022

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On the de-duplication of laion 2b

Foundation models

- Large players have different positioning
- Meta has access to nearly unlimited textual, visual data and behavioral provided by its users
- Alphabet has access to data from platforms such as YouTube and to behavioral data
- OpenAI has access to Microsoft owned data but

Axel Springer's OpenAI deal sets new template for media ties with Big Tech

Publishers want to avoid repeat of early internet era when US giants built ad-based empires using freely available content

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

- Good perspectives for Intellectual Property lawyers

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7 New York Times Says OpenAI Erased Potential Lawsuit Evidence

M As part of an ongoing copyright lawsuit, The New York Times says it spent 150
ch hours sifting through OpenAI's training data looking for potential evidence—
only for OpenAI to delete all of its work.

- Good perspectives for Intellectual Property lawyers

Comparative analysis of AI regulation - recommendations

China

- Delineate responsibility among government agencies
- Vision of AI ethics implementation
- Introduce a risk-based approach to AI development
- Ensure compliance enforcement

European Union

- Promoting domestic technologies and controlling them
- Importance given to systemic risks
- Increase transparency and control in recommender systems

Both

- Improve public engagement when formulating policy to align with citizens' interests
 - Control the influence of lobbying groups
 - Develop citizen assemblies on AI to represent the diversity of the population, trained by domain experts

Course projects

Motivation and objectives

- Hands-on experience with fairness-related topics
- Research-oriented approach
 - Full pipeline between idea and writing a (short) preprint
 - Intermediate presentations focused on: state of the art (C2), methodology (C3), experimental setup (C4), preliminary results (C5)
 - Final presentation during the last course
- All resources allowed but your personal touch is appreciated
- Group work - random selection of members
- Optional: continue working to submit an actual research article

Topics

1. Political domain representations in Grokipedia and Wikipedia
2. Test-time political bias reduction in LLMs
3. LLM permeability to falsehood, with focus on conspiracies
4. Evaluation of LLM biases for socially desirable political objectives
5. New source stance dynamics after owner change
6. Evolution of LLM political bias through time

Common points across projects

- Focus on entity bias by varying evaluation contexts
- Common set of politicians for all projects
- Work on small datasets to keep the projects feasible
- Focus on “small” LLMs - 7-8B parameters

Thank you!
Questions?

Today's assignment

Written essay (~30 minutes)

As future experts, you will push toward the adoption of AI technologies. Imagine that you discuss with a stakeholder who is skeptical about the role of ethics and regulation in AI, and you should argue against this view. Write up to one page of text countering the arguments given below.

The role of the stakeholder is played by ChatGPT 5.2 prompted with:

You are an artificial intelligence expert who is very skeptical about the validity of the European regulation-based approach meant to push toward the development of responsible and ethical AI. Provide arguments for your view, including factual examples of successful AI development in other regions of the world that choose to regulate less. The text should be approximately half a page long.

You should not use chatbots or other online resources during this exercise.

Europe's regulation-first strategy for "responsible AI" risks confusing *paper compliance* with *actual safety*. The EU AI Act is a comprehensive, risk-based regime with phased obligations (e.g., bans applying from February 2025; general-purpose AI obligations applying from August 2025; major high-risk rules applying from August 2026–2027). In practice, that kind of framework can (1) raise fixed compliance costs that hit SMEs hardest, (2) incentivize risk-averse legal defensibility over engineering robustness, and (3) push talent and scaling efforts to jurisdictions where iteration is faster. When the frontier shifts every few months, prescriptive requirements can also ossify yesterday's best practices—creating a "tick-the-box" culture rather than a continuous red-teaming / monitoring culture.

Meanwhile, some of the most visible AI advances of the last few years have come from ecosystems that leaned more on *voluntary standards, sectoral oversight, and ex post enforcement* than on a single horizontal AI law. In the United States, governance has been comparatively fragmented: NIST's AI Risk Management Framework is explicitly intended for voluntary use. Yet the U.S. environment produced rapid cycles of frontier model development and deployment—e.g., OpenAI's GPT-4 and GPT-4o releases—alongside extensive private-sector safety work (evals, system cards, staged rollouts). It also fostered well-capitalized competitors like Anthropic (Claude), which has scaled quickly on enterprise demand. And the enabling hardware/software stack has been dominated by U.S.-based firms such as NVIDIA, whose GPU platform underpins much of modern AI training and inference.

Other "lighter-touch" models show similar dynamics. The UK has explicitly framed a "pro-innovation" approach that relies on existing regulators applying cross-sector principles rather than creating a single new AI regulator-by-default. Singapore has emphasized practical, voluntary governance tooling (e.g., Model AI Governance guidance for GenAI and the open, test-oriented AI Verify ecosystem). The uncomfortable takeaway: ethics and responsibility don't automatically improve because a region writes detailed rules. They improve when incentives reward real-world risk reduction—measurable evaluations, incident reporting, liability for harm, and strong professional norms—while preserving the freedom to iterate quickly on methods that actually work.